Inducing the mother of all continued fractions

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Regular continued fractions and convergents

Denote by

$$x = [0; a_1, a_2, \dots] := \cfrac{1}{a_1 + \cfrac{1}{a_2 + \ddots}}$$

the regular continued fraction (r.c.f.)-expansion of $x \in (0,1)\backslash\mathbb{Q}$. Set $p_{-1} = q_0 := 1$, $p_0 = q_{-1} := 0$, and for positive $n \in \mathbb{N}$, let $p_n, q_n \in \mathbb{N}$ be the unique relatively prime integers satisfying

$$p_n/q_n = [0; a_1, \dots, a_n] := \cfrac{1}{a_1 + \cfrac{1}{\ddots + a_n}}$$

Call $p_n/q_n \in \mathbb{Q} \cup \{\infty\}$, $n \geq -1$, the nth *convergent* of x.

Recurrence relations and mediant convergents

The convergents of $x = [0; a_1, a_2, \dots]$ satisfy the following recurrence relations for all $n \geq 0$:

$$p_{n+1} = a_{n+1}p_n + p_{n-1}, \qquad p_{-1} = 1, \qquad p_0 = 0,$$
$$q_{n+1} = a_{n+1}q_n + q_{n-1}, \qquad q_{-1} = 0, \qquad q_0 = 1.$$

The *mediant convergents* of x are

$$\frac{\lambda p_n + p_{n-1}}{\lambda q_n + q_{n-1}}, \quad \lambda \in \{1, 2, \dots, a_{n+1} - 1\}.$$

Approximation coefficients

Definition

For $x \in (0,1)\backslash\mathbb{Q}$ and $p/q \in \mathbb{Q}$ with $(p,q) = 1$, the *approximation coefficient* $\Theta(x, p/q) \in \mathbb{R}$ of x and p/q is defined by

$$\left| x - \frac{p}{q} \right| = \frac{\Theta\left(x, p/q\right)}{q^2}.$$

That is,

$$\Theta\left(x, p/q\right) := q|qx - p|.$$

Legendre-type theorems

Theorem (Legendre 1798)

If $\Theta(x, p/q) < 1/2$, *then* p/q *is a convergent of* x. *Moreover, the constant* $1/2$ *is optimal.*

Theorem (Koksma 1937/38)

If $\Theta(x, p/q) < 2/3$, *then* p/q *is either a convergent or a first mediant convergent of* x. *The constant* $2/3$ *is optimal.*

Theorem (Fatou–Grace 1904/1918)

If $\Theta(x, p/q) < 1$, *then* p/q *is either a convergent or a nearest (first or final) mediant convergent of* x. *The constant* 1 *is optimal.*

(Partial) Converses and refinements

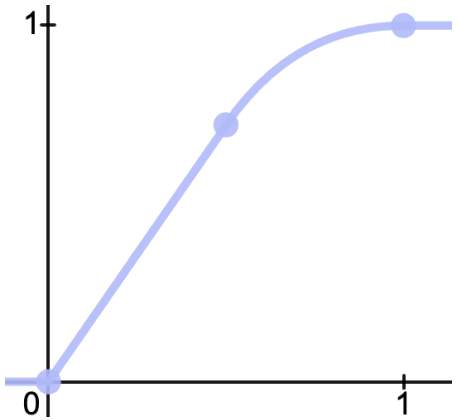
- $\Theta(x, p_n/q_n) < 1,\ n \geq 0$
- Kuipers–Meulenbeld 1952: Criteria on digits a_n to guarantee nearest mediants p/q satisfy $\Theta(x, p/q) < 1$.
- Barbolosi–Jager 1994: Refinements to Legendre, Fatou–Grace, Koksma and converses via assumptions on (the signature) of p/q.

Doeblin–Lenstra

In the 1980's Lenstra conjectured[1] that for a.e. $x \in (0,1)$ and all $z \in [0,1]$,

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \le k < n \mid \Theta(x, p_k/q_k) \le z\} = \begin{cases} \frac{1}{\log 2} z, & 0 \le z \le 1/2, \\ \frac{1}{\log 2} \left(1 - z + \log(2z)\right), & 1/2 \le z \le 1. \end{cases}$$

This was proven by Bosma, Jager and Wiedijk in 1983.



[1] This was also conjectured (proven?) by Doeblin in 1940.

Other limiting distributions

Limiting distributions corresponding to other sequences of convergents and mediants have been considered. These include:

- convergents and nearest mediants (Ito 1989, Bosma 1990, Jager 1991),
- convergents and k^{th}-nearest mediants (Brown–Yin 1996),
- λ^{th}-mediant convergents (Bosma 1990),
- final mediants (Bosma 1990),
- final mediants corresponding to a fixed digit (Bosma 1990),
- convergents/mediants of bounded approximation coefficients (Bosma 1990).

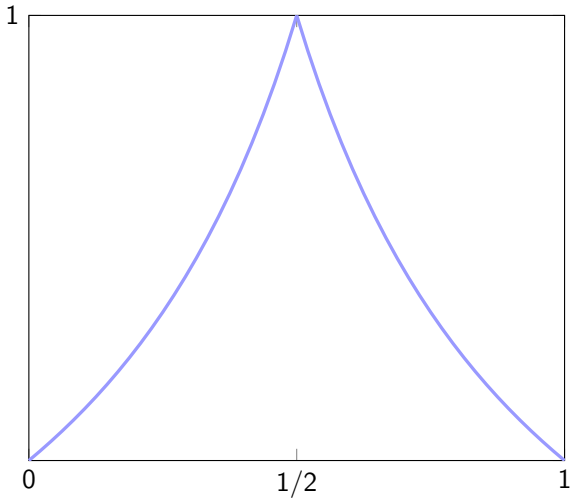
Farey tent map

Let $\varepsilon : [0,1] \to \{0,1\}$ be given by

$$\varepsilon(x) := \begin{cases} 0, & x \in [0, 1/2], \\ 1, & x \in (1/2, 1], \end{cases}$$

and for $\varepsilon \in \{0,1\}$, define

$$A_\varepsilon := \begin{pmatrix} 1-\varepsilon & \varepsilon \\ 1 & 1 \end{pmatrix} = \begin{cases} \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right), & \varepsilon = 0, \\ \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right), & \varepsilon = 1. \end{cases}$$



The *Farey tent map* $F : [0,1] \to [0,1]$ is

$$F(x) = A^{-1}_{\varepsilon(x)} \cdot x = \begin{cases} \frac{x}{1-x}, & \varepsilon(x) = 0, \\ \frac{1-x}{x}, & \varepsilon(x) = 1. \end{cases}$$

The system $([0,1], \mathcal{B}, \mu, F)$ is ergodic, where μ is the infinite, σ-finite, a.c. invariant measure with density $1/x$.

A maternal interlude

The Farey tent map F is conjugate via $x \mapsto x + 1$ to a map used by Lehner (1994) to produce semi-r.c.f.-expansions.

The Lehner expansion of x was termed the 'mother of all continued fractions' by Dajani–Kraaikamp (2000), as one may obtain from it all semi-r.c.f.-expansions of x via insertions/singularisations.

Dynamics of F

For $n \geq 0$, set $x_n := F^n(x)$ and $\varepsilon_{n+1} := \varepsilon(x_n)$. Then for $n \geq 1$,

$$x_n = A_{\varepsilon_n}^{-1} \cdots A_{\varepsilon_2}^{-1} A_{\varepsilon_1}^{-1} \cdot x = (A_{\varepsilon_1} A_{\varepsilon_2} \cdots A_{\varepsilon_n})^{-1} \cdot x.$$

Set

$$A_{[0,n]} = A_{[0,n]}(x) := A_{\varepsilon_1} A_{\varepsilon_2} \cdots A_{\varepsilon_n}, \quad n \geq 0,$$

so that $x_n = A_{[0,n]}^{-1} \cdot x$.

Dynamics of F (cont'd)

Suppose $x = [0; a_1, a_2, \dots] \notin \mathbb{Q}$. If $a_1 > 1$, then

$$F(x) = \frac{x}{1-x} = \frac{1}{\frac{1}{x} - 1} = [0; a_1 - 1, a_2, \dots],$$

while if $a_1 = 1$,

$$F(x) = \frac{1-x}{x} = \frac{1}{x} - 1 = [0; a_2, a_3, \dots].$$

That is,

$$F(x) = \begin{cases} [0; a_1 - 1, a_2, \dots], & a_1 > 1, \\ [0; a_2, a_3, \dots], & a_1 = 1, \end{cases}$$

and thus

$$\varepsilon_1\varepsilon_2\varepsilon_3\cdots = 0^{a_1-1}10^{a_2-1}10^{a_3-1}1\cdots .$$

Matrices $A_{[0,n]}$

For $n \geq 0$, let $j_n = j_n(x)$ and $\lambda_n = \lambda_n(x)$ be the unique integers satisfying

$$n = a_1 + a_2 + \cdots + a_{j_n} + \lambda_n, \quad j_n \geq 0, \quad 0 \leq \lambda_n < a_{j_n+1}.$$

we find

$$\begin{pmatrix} u_n & t_n \\ s_n & r_n \end{pmatrix} := A_{[0,n]} = A_{\varepsilon_1} A_{\varepsilon_2} \cdots A_{\varepsilon_n}$$

$$= A_0^{a_1-1} A_1 \cdots A_0^{a_{j_n}-1} A_1 A_0^{\lambda_n}$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & a_1 \end{pmatrix} \cdots \begin{pmatrix} 0 & 1 \\ 1 & a_{j_n} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \lambda_n & 1 \end{pmatrix}$$

$$= \begin{pmatrix} p_{j_n-1} & p_{j_n} \\ q_{j_n-1} & q_{j_n} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \lambda_n & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \lambda_n p_{j_n} + p_{j_n-1} & p_{j_n} \\ \lambda_n q_{j_n} + q_{j_n-1} & q_{j_n} \end{pmatrix}.$$

$$A_0^{a-1} A_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{a-1} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ a-1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & a \end{pmatrix}$$

F-orbits and convergents/mediants

$$A_{[0,n]} = \begin{pmatrix} u_n & t_n \\ s_n & r_n \end{pmatrix} = \begin{pmatrix} \lambda_n p_{j_n} + p_{j_n-1} & p_{j_n} \\ \lambda_n q_{j_n} + q_{j_n-1} & q_{j_n} \end{pmatrix}$$

Via $x_n = A_{[0,n]}^{-1} \cdot x$, we find a natural 1-1 correspondence between points in the orbit $(x_n)_{n\geq 0}$ and the r.c.f. convergents and mediants $(u_n/s_n)_{n\geq 0}$ of x. Note the order:

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(A_{[0,n]} \cdot \infty\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0+p_{-1}}{q_0+q_{-1}}, \ldots, \frac{(a_1-1)p_0+p_{-1}}{(a_1-1)q_0+q_{-1}}, \right.$$
$$\frac{p_0}{q_0}, \frac{p_1+p_0}{q_1+q_0}, \ldots, \frac{(a_2-1)p_1+p_0}{(a_2-1)q_1+q_0}, \ldots,$$
$$\left.\frac{p_{k-1}}{q_{k-1}}, \frac{p_k+p_{k-1}}{q_k+q_{k-1}}, \ldots, \frac{(a_{k+1}-1)p_k+p_{k-1}}{(a_{k+1}-1)q_k+q_{k-1}}, \ldots\right).$$

Ito's 'remedy'

$$A_{[0,n]} = \begin{pmatrix} u_n & t_n \\ s_n & r_n \end{pmatrix} = \begin{pmatrix} \lambda_n p_{j_n} + p_{j_n-1} & p_{j_n} \\ \lambda_n q_{j_n} + q_{j_n-1} & q_{j_n} \end{pmatrix}$$

Ito (1989) considers instead

$$\left(\frac{u_n + t_n}{s_n + r_n}\right)_{n\geq 0} = \left(A_{[0,n]} \cdot 1\right)_{n\geq 0}$$

$$= \left(\frac{p_0 + p_{-1}}{q_0 + q_{-1}}, \frac{2p_0 + p_{-1}}{2q_0 + q_{-1}}, \ldots, \frac{(a_1 - 1)p_0 + p_{-1}}{(a_1 - 1)q_0 + q_{-1}}, \frac{p_1}{q_1}, \right.$$
$$\frac{p_1 + p_0}{q_1 + q_0}, \frac{2p_1 + p_0}{2q_1 + q_0}, \ldots, \frac{(a_2 - 1)p_1 + p_0}{(a_2 - 1)q_1 + q_0}, \frac{p_2}{q_2}, \ldots,$$
$$\left. \frac{p_k + p_{k-1}}{q_k + q_{k-1}}, \frac{2p_k + p_{k-1}}{2q_k + q_{k-1}}, \ldots, \frac{(a_{k+1} - 1)p_k + p_{k-1}}{(a_{k+1} - 1)q_k + q_{k-1}}, \frac{p_{k+1}}{q_{k+1}}, \ldots \right).$$

Ito's natural extension

Set $\Omega := [0,1]^2$ and define $\mathcal{F} : \Omega \to \Omega$ by

$$\mathcal{F}(x,y) := \left(A^{-1}_{\varepsilon(x)} \cdot x, A_{\varepsilon(x)} \cdot y\right) = \begin{cases} \left(\frac{x}{1-x}, \frac{y}{1+y}\right), & x \in [0,1/2], \\ \left(\frac{1-x}{x}, \frac{1}{1+y}\right), & x \in (1/2,1]. \end{cases}$$

For $(x,y) = ([0; a_1, a_2, \ldots], [0; b_1, b_2, \ldots]) \in \Omega$,

$$\mathcal{F}(x,y) = \begin{cases} ([0; a_1 - 1, a_2, \ldots], [0; b_1 + 1, b_2, \ldots]), & a_1 > 1, \\ ([0; a_2, a_3 \ldots], [0; 1, b_1, b_2, \ldots]), & a_1 = 1. \end{cases}$$

Ito (1989) showed that $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ is the natural extension of $([0,1], \mathcal{B}, \mu, F)$, where $\bar{\mu}$ is the infinite, σ-finite, a.c. invariant measure with density $1/(x+y-xy)^2$. Ergodicity of F implies that of $\mathcal{F}$.

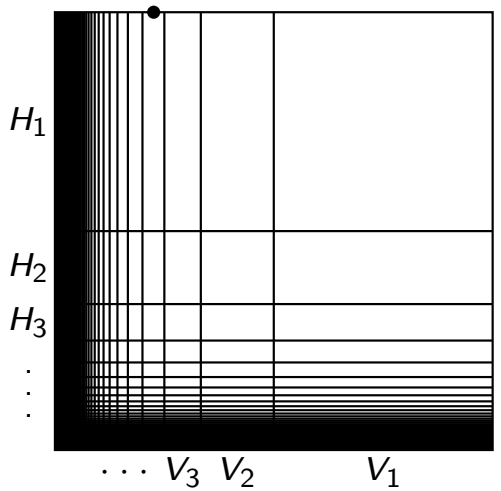
Dynamics of $\mathcal{F}$

$$\mathcal{F}(x,y) = \begin{cases} ([0;a_1-1,a_2,\dots],[0;b_1+1,b_2,\dots]), & a_1 > 1, \\ ([0;a_2,a_3\dots],[0;1,b_1,b_2,\dots]), & a_1 = 1. \end{cases}$$

Example

Let $(x,y) = ([0;\overline{4,2,3}],[0;1])$. We have

$$\begin{aligned}
(x,y) &= ([0;4,\overline{2,3,4}],[0;1]) \\
\mathcal{F}(x,y) &= ([0;3,\overline{2,3,4}],[0;2]) \\
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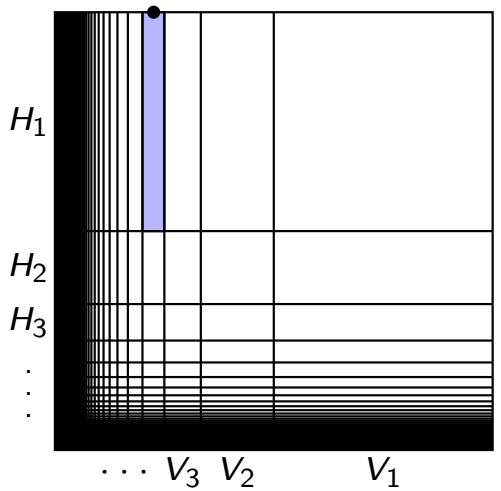
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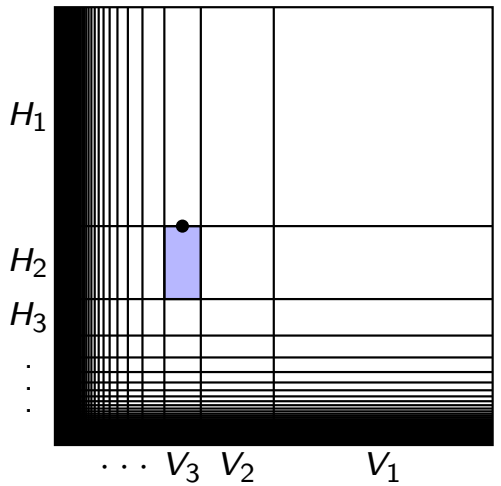
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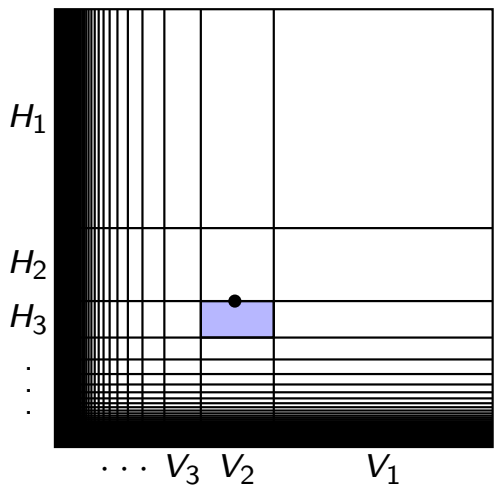
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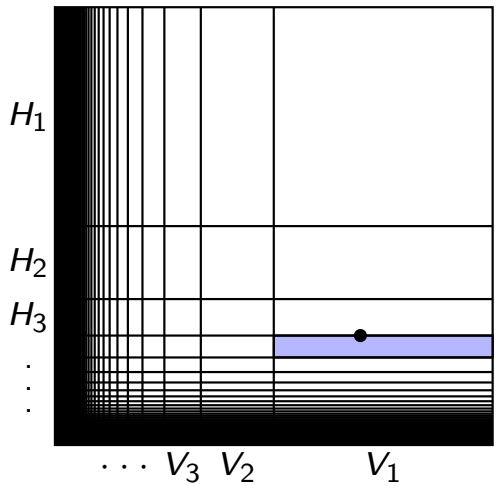
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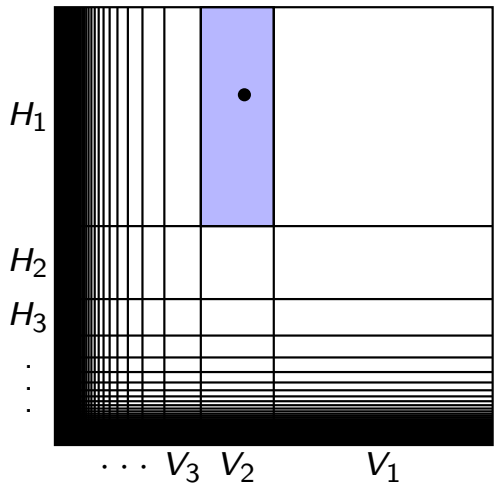
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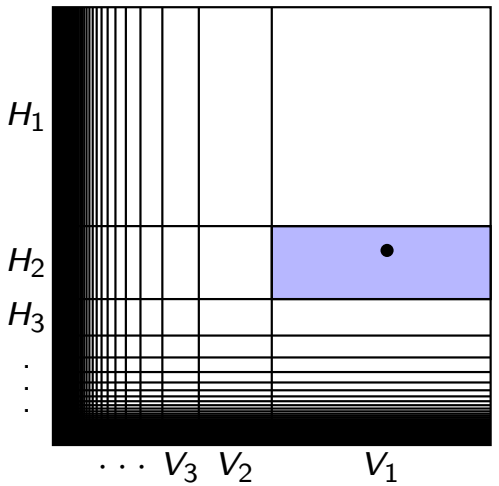
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$$\mathcal{F}(x,y) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3 \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

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\mathcal{F}^2(x,y) &= ([0; 2, \overline{2, 3, 4}], [0; 3]) \\
\mathcal{F}^3(x,y) &= ([0; 1, \overline{2, 3, 4}], [0; 4]) \\
\mathcal{F}^4(x,y) &= ([0; 2, \overline{3, 4, 2}], [0; 1, 4]) \\
\mathcal{F}^5(x,y) &= ([0; 1, \overline{3, 4, 2}], [0; 2, 4]) \\
\mathcal{F}^6(x,y) &= ([0; 3, \overline{4, 2, 3}], [0; 1, 2, 4]) \\
\mathcal{F}^7(x,y) &= ([0; 2, \overline{4, 2, 3}], [0; 2, 2, 4]) \\
\mathcal{F}^8(x,y) &= ([0; 1, \overline{4, 2, 3}], [0; 3, 2, 4])
\end{aligned}$$



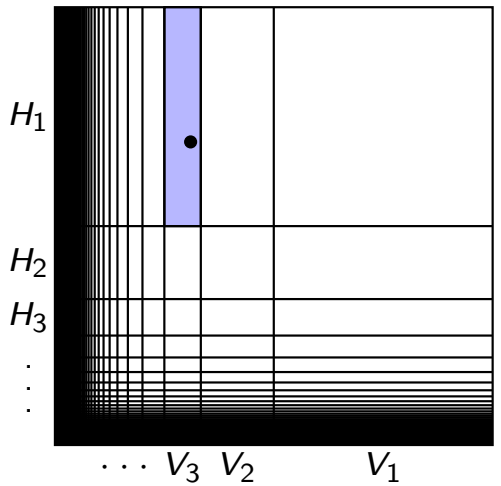
Dynamics of $\mathcal{F}$

$$\mathcal{F}(x,y) = \begin{cases} ([0; a_1 - 1, a_2, \dots], [0; b_1 + 1, b_2, \dots]), & a_1 > 1, \\ ([0; a_2, a_3 \dots], [0; 1, b_1, b_2, \dots]), & a_1 = 1. \end{cases}$$

Example

Let $(x, y) = ([0; \overline{4, 2, 3}], [0; 1])$. We have

$$\begin{aligned}
(x,y) &= ([0; 4, \overline{2, 3, 4}], [0; 1]) \\
\mathcal{F}(x,y) &= ([0; 3, \overline{2, 3, 4}], [0; 2]) \\
\mathcal{F}^2(x,y) &= ([0; 2, \overline{2, 3, 4}], [0; 3]) \\
\mathcal{F}^3(x,y) &= ([0; 1, \overline{2, 3, 4}], [0; 4]) \\
\mathcal{F}^4(x,y) &= ([0; 2, \overline{3, 4, 2}], [0; 1, 4]) \\
\mathcal{F}^5(x,y) &= ([0; 1, \overline{3, 4, 2}], [0; 2, 4]) \\
\mathcal{F}^6(x,y) &= ([0; 3, \overline{4, 2, 3}], [0; 1, 2, 4]) \\
\mathcal{F}^7(x,y) &= ([0; 2, \overline{4, 2, 3}], [0; 2, 2, 4]) \\
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\end{aligned}$$



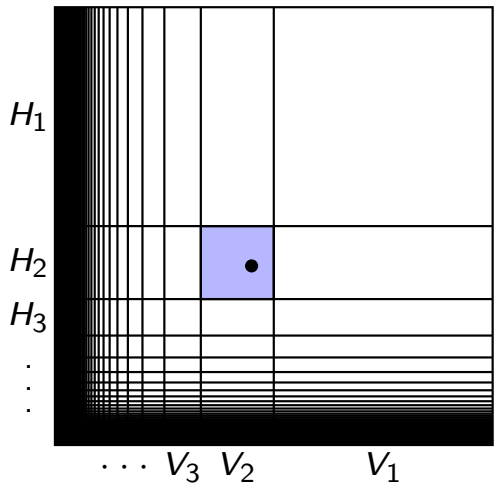
Dynamics of $\mathcal{F}$

$$\mathcal{F}(x,y) = \begin{cases} ([0;a_1-1,a_2,\dots],[0;b_1+1,b_2,\dots]), & a_1 > 1, \\ ([0;a_2,a_3\dots],[0;1,b_1,b_2,\dots]), & a_1 = 1. \end{cases}$$

Example

Let $(x,y) = ([0;\overline{4,2,3}],[0;1])$. We have

$$\begin{aligned}
(x,y) &= ([0;4,\overline{2,3,4}],[0;1]) \\
\mathcal{F}(x,y) &= ([0;3,\overline{2,3,4}],[0;2]) \\
\mathcal{F}^2(x,y) &= ([0;2,\overline{2,3,4}],[0;3]) \\
\mathcal{F}^3(x,y) &= ([0;1,\overline{2,3,4}],[0;4]) \\
\mathcal{F}^4(x,y) &= ([0;2,\overline{3,4,2}],[0;1,4]) \\
\mathcal{F}^5(x,y) &= ([0;1,\overline{3,4,2}],[0;2,4]) \\
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\mathcal{F}^8(x,y) &= ([0;1,\overline{4,2,3}],[0;3,2,4])
\end{aligned}$$



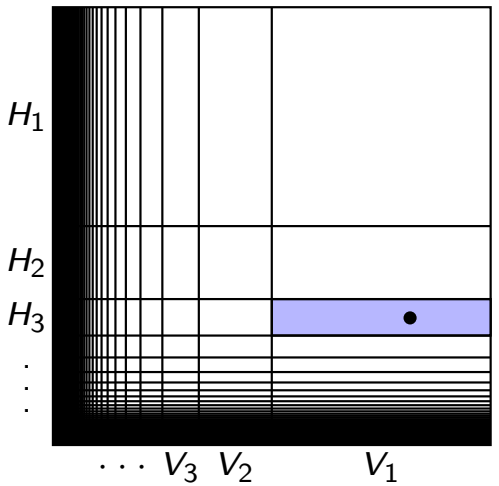
Dynamics of $\mathcal{F}$

$$\mathcal{F}(x,y) = \begin{cases} ([0;a_1-1,a_2,\dots],[0;b_1+1,b_2,\dots]), & a_1 > 1, \\ ([0;a_2,a_3\dots],[0;1,b_1,b_2,\dots]), & a_1 = 1. \end{cases}$$

Example

Let $(x,y) = ([0;\overline{4,2,3}],[0;1])$. We have

$$\begin{aligned}
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\end{aligned}$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

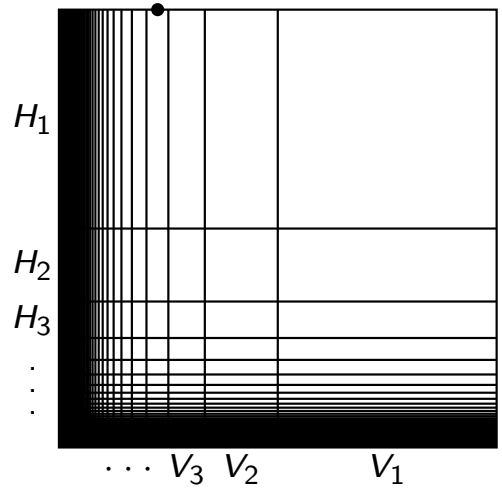
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_0, y_0) = ([0; 4, \overline{2,3,4}], [0;1]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0+p_{-1}}{p_0+q_{-1}}, \frac{2q_0+p_{-1}}{2q_0+q_{-1}}, \frac{3p_0+p_{-1}}{3q_0+q_{-1}}, \right.$$
$$\frac{p_0}{q_0}, \frac{p_1+p_0}{q_1+q_0},$$
$$\left. \frac{p_1}{q_1}, \frac{p_2+p_1}{q_2+q_1}, \frac{2p_2+p_1}{2q_2+q_1}, \dots \right)$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

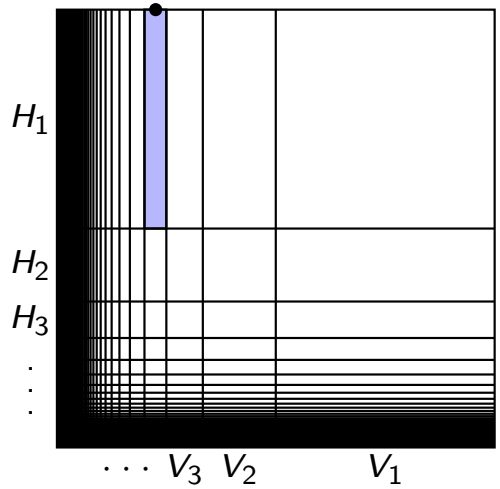
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_0, y_0) = ([0; 4, \overline{2,3,4}], [0;1]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
$$\frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0},$$
$$\left. \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \cdots \right)$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

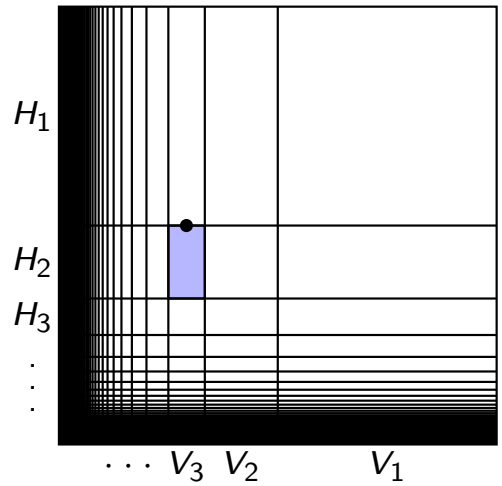
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_1, y_1) = ([0; 3, \overline{2,3,4}], [0;2]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
$$\frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0},$$
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$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

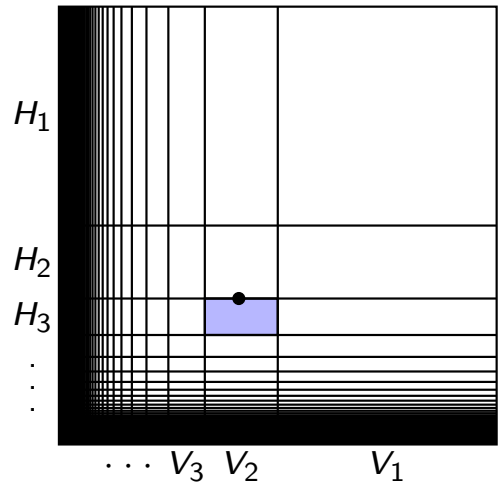
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_2, y_2) = ([0; 2, \overline{2,3,4}], [0;3]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
$$\frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0},$$
$$\left. \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \cdots \right)$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

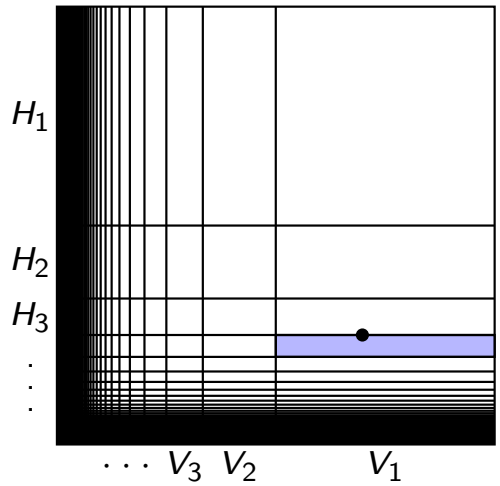
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_3, y_3) = ([0; 1, \overline{2,3,4}], [0; 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \right.$$
$$\frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0},$$
$$\left. \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \cdots \right)$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

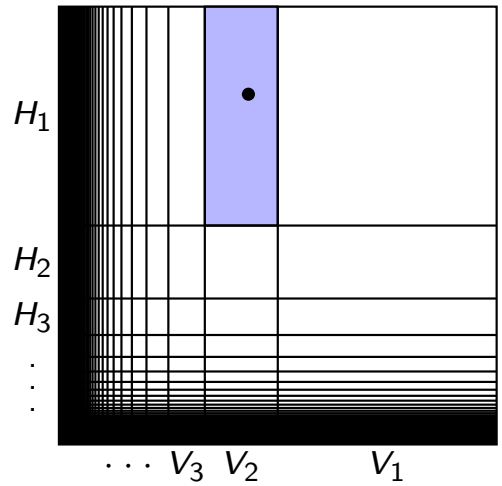
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_4, y_4) = ([0; 2, \overline{3,4,2}], [0; 1, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0}, \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \cdots\right)$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

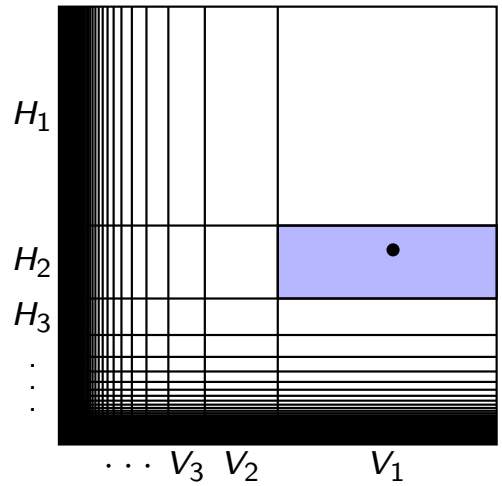
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_5, y_5) = ([0; 1, \overline{3,4,2}], [0; 2, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0}, \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \cdots\right)$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

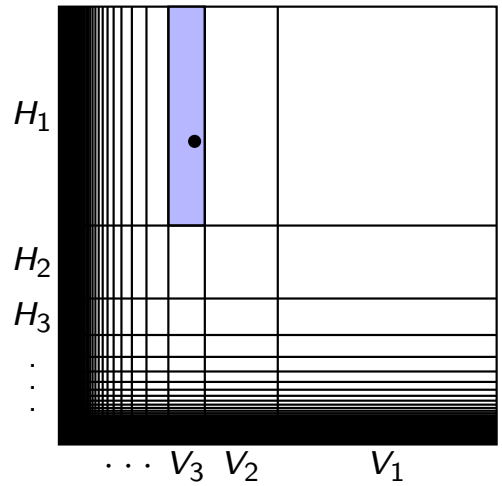
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_6, y_6) = ([0; 3, \overline{4,2,3}], [0; 1, 2, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0}, \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \cdots\right)$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

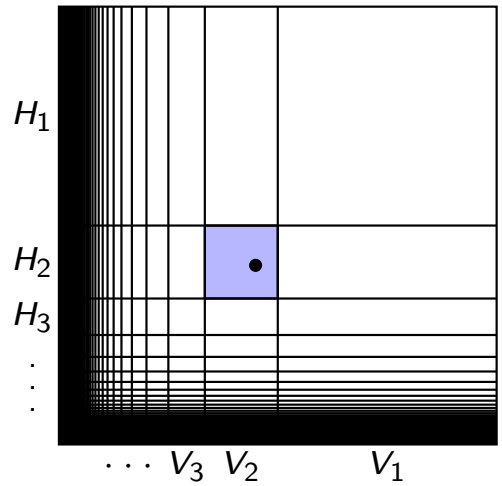
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_7, y_7) = ([0; 2, \overline{4,2,3}], [0; 2, 2, 4]),$$

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0}, \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \cdots\right)$$



$\mathcal{F}$-orbits and convergents/mediants

For $x \in (0,1)\backslash\mathbb{Q}$ and $n \geq 0$ let $(x_n, y_n) := \mathcal{F}^n(x,1)$. Extend the 1-1 correspondence between r.c.f. convergents/mediants and the F-orbit of x to the $\mathcal{F}$-orbit of x:

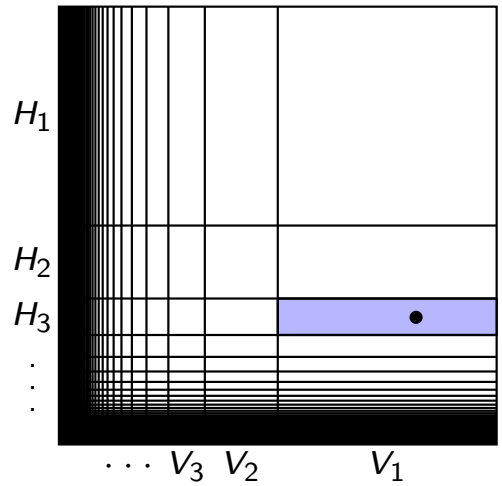
$$\frac{u_n}{s_n} = \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n q_{j_n} + q_{j_n-1}} \quad \longleftrightarrow \quad x_n \quad \longleftrightarrow \quad (x_n, y_n).$$

Example

Let $x = [0; \overline{4,2,3}]$. We have

$$(x_8, y_8) = ([0; 1, \overline{4,2,3}], [0; 3, 2, 4]),$$

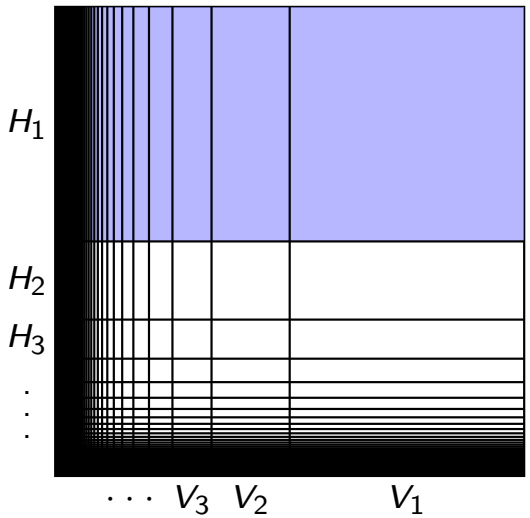
$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \left(\frac{p_{-1}}{q_{-1}}, \frac{p_0 + p_{-1}}{p_0 + q_{-1}}, \frac{2q_0 + p_{-1}}{2q_0 + q_{-1}}, \frac{3p_0 + p_{-1}}{3q_0 + q_{-1}}, \frac{p_0}{q_0}, \frac{p_1 + p_0}{q_1 + q_0}, \frac{p_1}{q_1}, \frac{p_2 + p_1}{q_2 + q_1}, \frac{2p_2 + p_1}{2q_2 + q_1}, \cdots\right)$$



Subregions of Ω and convergents/mediants

Note that particular subregions of Ω correspond to particular convergents/mediants:

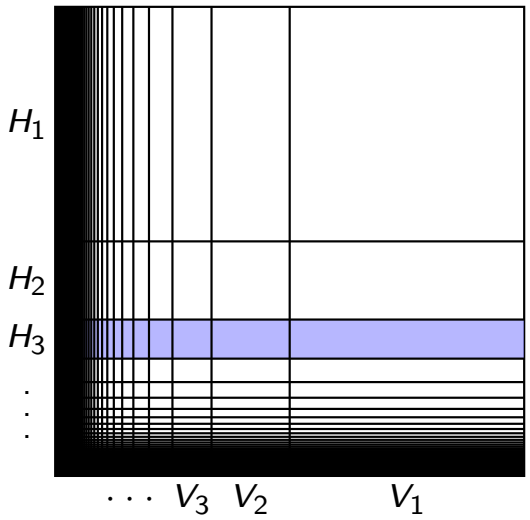
$$(x_n, y_n) \in H_1 \iff \frac{u_n}{s_n} = \frac{p_{j_n-1}}{q_{j_n-1}}$$



Subregions of Ω and convergents/mediants

Note that particular subregions of Ω correspond to particular convergents/mediants:

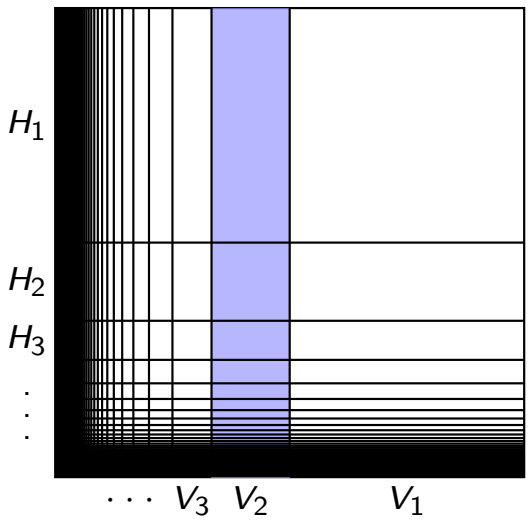
$$(x_n, y_n) \in H_{\lambda+1} \iff \frac{u_n}{s_n} = \frac{\lambda p_{j_n} + p_{j_n-1}}{\lambda q_{j_n} + q_{j_n-1}}, \quad a_{j_n+1} > \lambda$$



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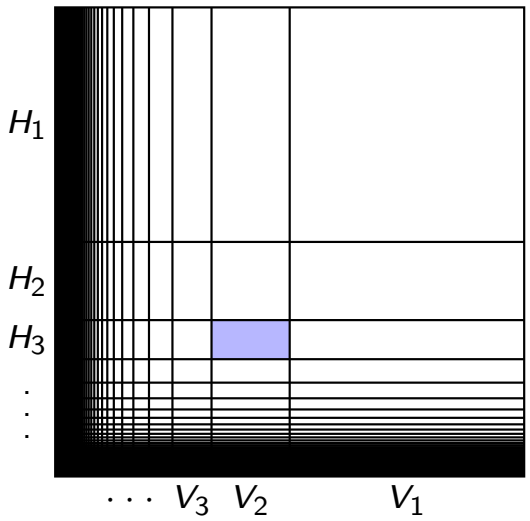
$$(x_n, y_n) \in V_a \iff \frac{u_n}{s_n} = \frac{(a_{j_n+1} - a)p_{j_n} + p_{j_n-1}}{(a_{j_n+1} - a)q_{j_n} + q_{j_n-1}}, \quad a_{j_n+1} \geq a$$



Subregions of Ω and convergents/mediants

Note that particular subregions of Ω correspond to particular convergents/mediants:

$$(x_n, y_n) \in V_{a-\lambda} \cap H_{\lambda+1} \iff \frac{u_n}{s_n} = \frac{\lambda p_{j_n} + p_{j_n-1}}{\lambda q_{j_n} + q_{j_n-1}}, \quad a_{j_n+1} = a$$



Induced systems

Fix $R \subset \Omega$ with $0 < \bar{\mu}(R) < \infty$ and $\bar{\mu}(\partial R) = 0$. Let $\mathcal{F}_R : R \cup ([0,1] \times \{1\}) \to R$ be $\mathcal{F}$ 'induced' on R; that is,

$$\mathcal{F}_R(x,y) := \mathcal{F}^{r_R(x,y)}(x,y),$$

where $r_R : \Omega \to \mathbb{N} \cup \{\infty\}$ denotes the hitting time to R, i.e.,

$$r_R(x,y) := \inf\{\ n \geq 1 \mid \mathcal{F}^n(x,y) \in R\}.$$

Denote by $\bar{\mu}_R$ the induced (probability) measure:

$$\bar{\mu}_R(S) = \frac{\bar{\mu}(S)}{\bar{\mu}(R)}, \qquad S \in \mathcal{B} \cap R := \{B \cap R \mid B \in \mathcal{B}\}.$$

Since $(\Omega, \mathcal{B}, \bar{\mu}, \mathcal{F})$ is ergodic, so is $(R, \mathcal{B} \cap R, \bar{\mu}_R, \mathcal{F}_R)$.

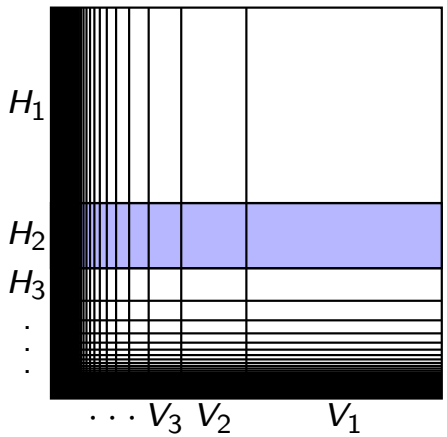
$\mathcal{F}_R$-orbits and mediants/convergents

The region R naturally determines a subsequence $((x_n^R, y_n^R))_{n\geq 0} := (\mathcal{F}_R^n(x,1))_{n\geq 0}$ of $((x_n, y_n))_{n\geq 0}$ and hence also a subsequence $(u_n^R/s_n^R)_{n\geq 0}$ of $(u_n/s_n)_{n\geq 0}$ via $(x_n, y_n) \longleftrightarrow u_n/s_n$.

Example

Let $R = H_2$. Then

$$\left(\frac{u_n}{s_n}\right)_{n\geq 0} = \Bigg(\frac{p_{-1}}{q_{-1}}, \frac{p_0+p_{-1}}{q_0+q_{-1}}, \ldots, \frac{(a_1-1)p_0+p_{-1}}{(a_1-1)q_0+q_{-1}},$$
$$\frac{p_0}{q_0}, \frac{p_1+p_0}{q_1+q_0}, \ldots, \frac{(a_2-1)p_1+p_0}{(a_2-1)q_1+q_0}, \ldots,$$
$$\frac{p_{k-1}}{q_{k-1}}, \frac{p_k+p_{k-1}}{q_k+q_{k-1}}, \ldots, \frac{(a_{k+1}-1)p_k+p_{k-1}}{(a_{k+1}-1)q_k+q_{k-1}}, \ldots\Bigg).$$



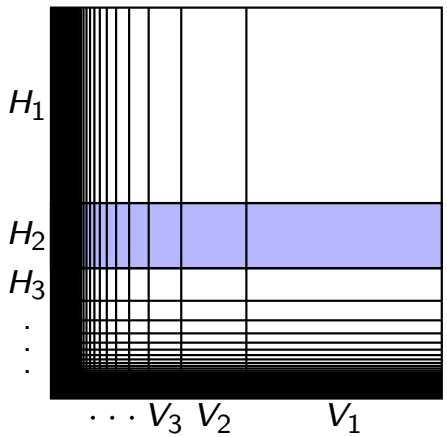
$\mathcal{F}_R$-orbits and mediants/convergents

The region R naturally determines a subsequence $((x_n^R, y_n^R))_{n\geq 0} := (\mathcal{F}_R^n(x,1))_{n\geq 0}$ of $((x_n, y_n))_{n\geq 0}$ and hence also a subsequence $(u_n^R/s_n^R)_{n\geq 0}$ of $(u_n/s_n)_{n\geq 0}$ via $(x_n, y_n) \longleftrightarrow u_n/s_n$.

Example

Let $R = H_2$. Then

$$\left(\frac{u_n}{s_n}^R\right)_{n\geq 0} = \Bigg(\frac{p_{-1}}{q_{-1}}, \frac{p_0+p_{-1}}{q_0+q_{-1}}, \ldots, \frac{(a_1-1)p_0+p_{-1}}{(a_1-1)q_0+q_{-1}},$$
$$\frac{p_0}{q_0}, \frac{p_1+p_0}{q_1+q_0}, \ldots, \frac{(a_2-1)p_1+p_0}{(a_2-1)q_1+q_0}, \ldots,$$
$$\frac{p_{k-1}}{q_{k-1}}, \frac{p_k+p_{k-1}}{q_k+q_{k-1}}, \ldots, \frac{(a_{k+1}-1)p_k+p_{k-1}}{(a_{k+1}-1)q_k+q_{k-1}}, \ldots\Bigg).$$



Approximation coefficients

Define $h : \Omega \to [0, \infty]$ by $h(x, y) = \frac{1-y}{x+y-xy}$.

Proposition

For any $n \geq 0$,

$$\Theta(x, u_n/s_n) = h(x_n, y_n) \quad \textit{and} \quad \Theta(x, u_n^R/s_n^R) = h(x_n^R, y_n^R).$$

Sketch.

Recall $A_{[0,n]} = \left(\begin{smallmatrix} u_n & t_n \\ s_n & r_n \end{smallmatrix}\right)$. Since $x = A_{[0,n]} \cdot x_n = \frac{u_n x_n + t_n}{s_n x_n + r_n}$,

$$\Theta(x, u_n/s_n) = s_n^2 \left| x - \frac{u_n}{s_n} \right| = s_n^2 \left| \frac{u_n x_n + t_n}{s_n x_n + r_n} - \frac{u_n}{s_n} \right| = \frac{s_n \left| t_n s_n - u_n r_n \right|}{s_n x_n + r_n} = \frac{s_n}{s_n x_n + r_n}.$$

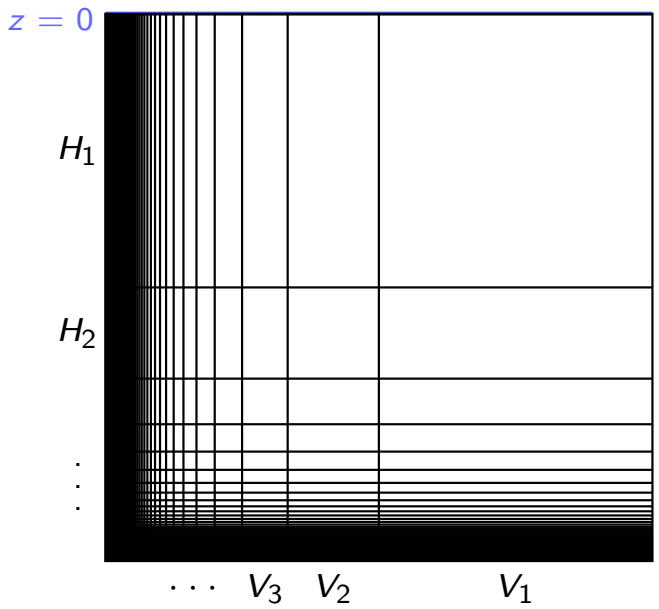
One shows that $y_n = r_n/(s_n + r_n)$, so

$$h(x_n, y_n) = \frac{1 - y_n}{x_n + y_n - x_n y_n} = \frac{1 - y_n}{(1 - y_n)x_n + y_n} = \frac{s_n}{s_n x_n + r_n}.$$

Approximation coefficients (cont'd)

Thus for $x = [0; a_1, a_2, \dots] \notin \mathbb{Q}$, $n \geq 0$ and $z \in [0, \infty]$,

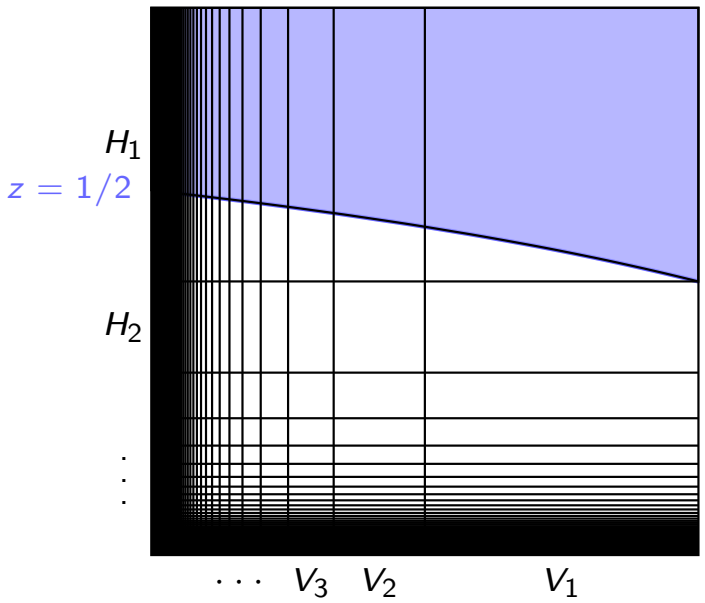
$$\Theta\left(x, \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n p_{j_n} + p_{j_n - 1}}\right) = \Theta\left(x, \frac{u_n}{s_n}\right) < z \quad \Longleftrightarrow \quad h(x_n, y_n) < z.$$



Approximation coefficients (cont'd)

Thus for $x = [0; a_1, a_2, \dots] \notin \mathbb{Q}$, $n \geq 0$ and $z \in [0, \infty]$,

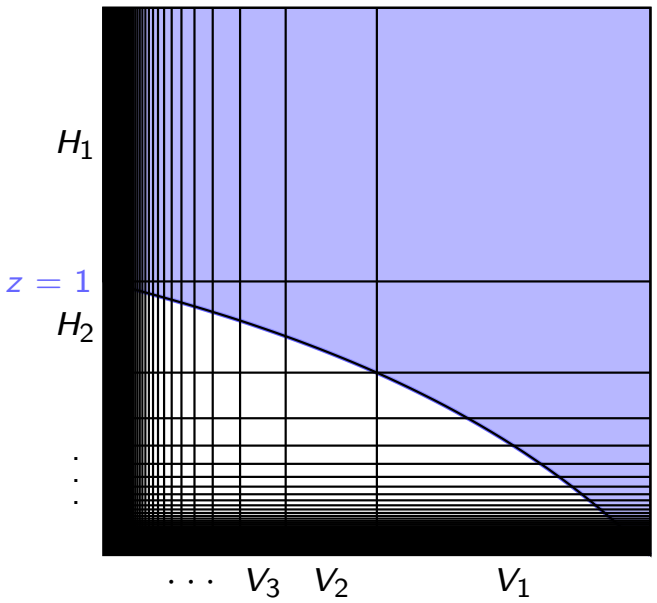
$$\Theta\left(x, \frac{\lambda_n p_{j_n} + p_{j_n-1}}{\lambda_n p_{j_n} + p_{j_n-1}}\right) = \Theta\left(x, \frac{u_n}{s_n}\right) < z \quad \Longleftrightarrow \quad h(x_n, y_n) < z.$$



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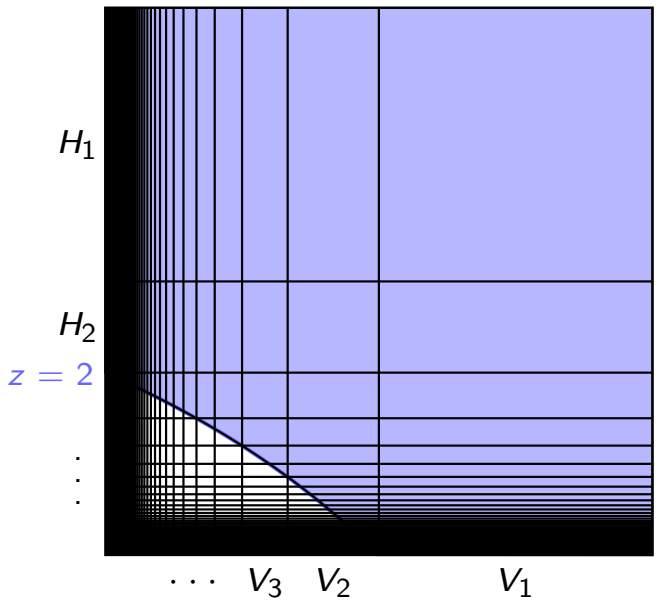
$$\Theta\left(x, \frac{\lambda_n p_{j_n} + p_{j_n - 1}}{\lambda_n p_{j_n} + p_{j_n - 1}}\right) = \Theta\left(x, \frac{u_n}{s_n}\right) < z \quad \Longleftrightarrow \quad h(x_n, y_n) < z.$$



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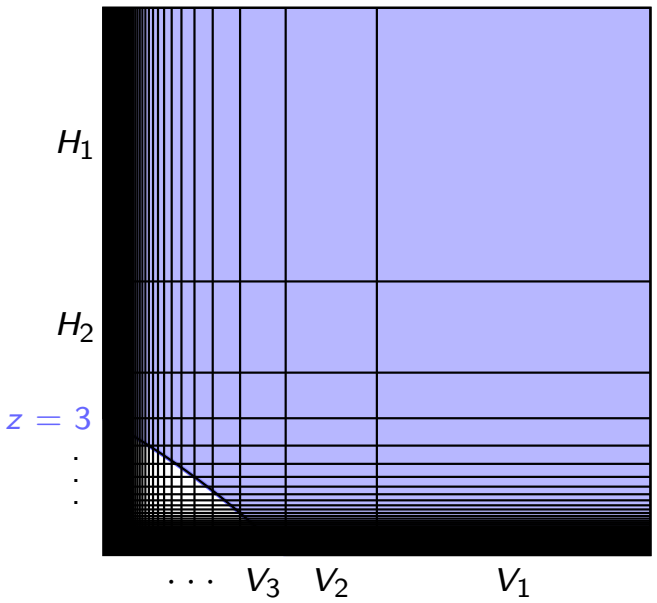
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Approximation coefficients (cont'd)

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Distribution of (x_n^R, y_n^R)

Proposition

For almost all $x \in (0,1)\backslash\mathbb{Q}$, the sequence (x_n^R, y_n^R) is distributed over R with density

$$\frac{1}{\bar{\mu}(R)(x+y-xy)^2}.$$

That is, for any $S \in \mathcal{B} \cap R$ with $\bar{\mu}(\partial S) = 0$,

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbf{1}_S(x_k^R, y_k^R) = \bar{\mu}_R(S) \quad \textit{almost surely.}$$

Distribution of bounded $\Theta(x, u_n^R/s_n^R)$'s

Set $S_z := \{(x,y) \in \Omega \mid h(x,y) \le z\},\ z \in [0,\infty)$.

Corollary

For any $R \subset \Omega$ with $0 < \bar{\mu}(R) < \infty,\ \bar{\mu}(\partial R) = 0$, any $z \in [0,\infty)$ and a.e. $x \in (0,1)$,

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \le k < n \mid \Theta(x, u_k^R/s_k^R) \le z\} = \bar{\mu}_R(S_z \cap R).$$

Proof.

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{n} \#\{0 \le k < n \mid \Theta(x, u_k^R/s_k^R) \le z\} &= \lim_{n\to\infty} \frac{1}{n} \#\{0 \le k < n \mid h(x_k^R, y_k^R) \le z\} \\ &= \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \mathbf{1}_{S_z \cap R}(x_k^R, y_k^R) \\ &= \bar{\mu}_R(S_z \cap R) \quad \text{a.s.}\end{aligned}$$

□

Convergents (Doeblin–Lenstra)

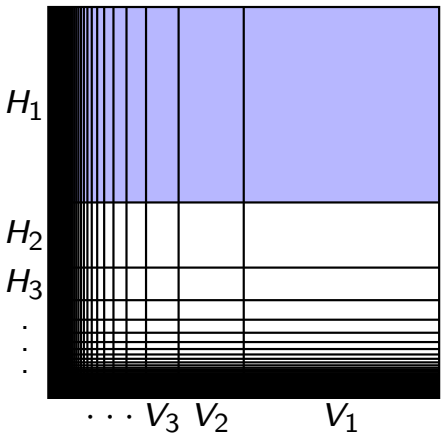
Theorem (Bosma–Jager–Wiedijk 1983)

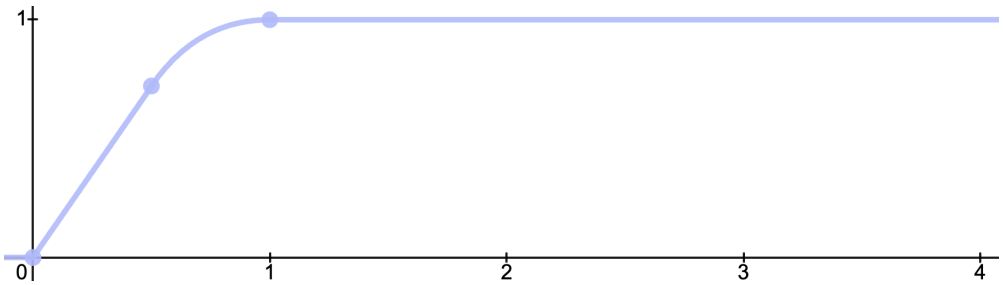
Set $R := H_1$. *Then*

$$\left\{ \frac{u_n^R}{s_n^R} \right\} = \left\{ \frac{p_{j-1}}{q_{j-1}} \right\}$$

and

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \le k < n \mid \Theta(x, u_k^R/s_k^R) \le z\} = \bar{\mu}_R(S_z \cap R) \quad \text{a.s.}$$





λ^{th} mediants

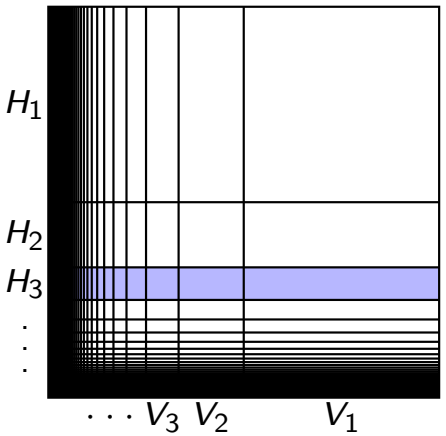
Theorem (Bosma 1990)

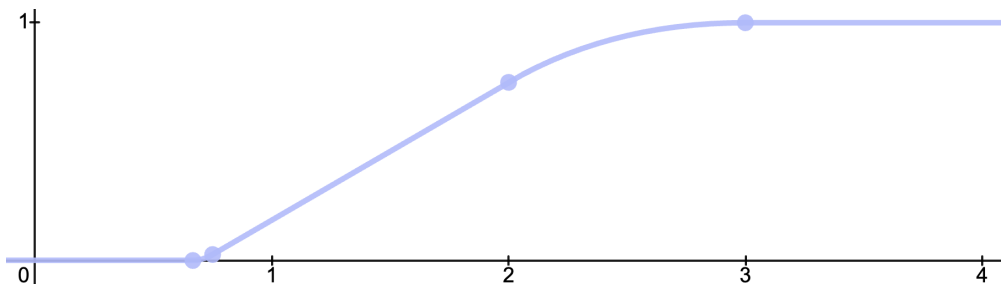
Set $R := H_{\lambda+1}$, $\lambda > 0$. *Then*

$$\left\{ \frac{u_n^R}{s_n^R} \right\} = \left\{ \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}} \right\}$$

and

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \le k < n \mid \Theta(x, u_k^R/s_k^R) \le z\} = \bar{\mu}_R(S_z \cap R) \quad \text{a.s.}$$





a^{th}-from-final mediants & $(a_{j+1} = a)$-convergents

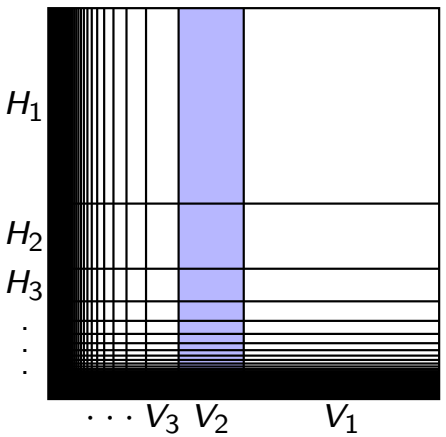
Theorem

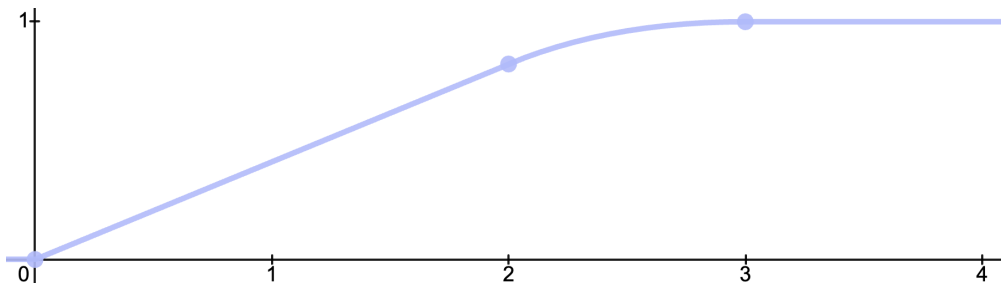
Set $R := V_a$, $a > 0$. *Then*

$$\left\{ \frac{u_n^R}{s_n^R} \right\} = \left\{ \frac{(a_{j+1} - a)p_j + p_{j-1}}{(a_{j+1} - a)q_j + q_{j-1}} \;\middle|\; a_{j+1} \geq a \right\}$$

and

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \leq k < n \mid \Theta(x, u_k^R/s_k^R) \leq z\} = \bar{\mu}_R(S_z \cap R) \quad \text{a.s.}$$





a^{th}-from-final mediants

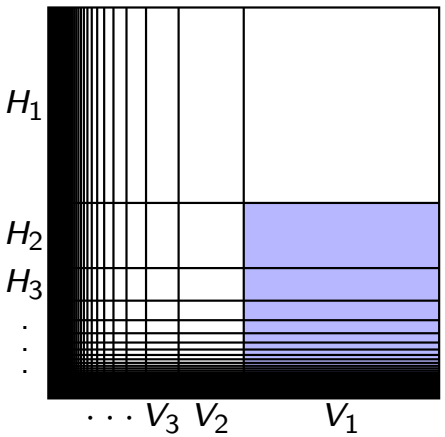
Theorem (Bosma 1990, $a = 1$)

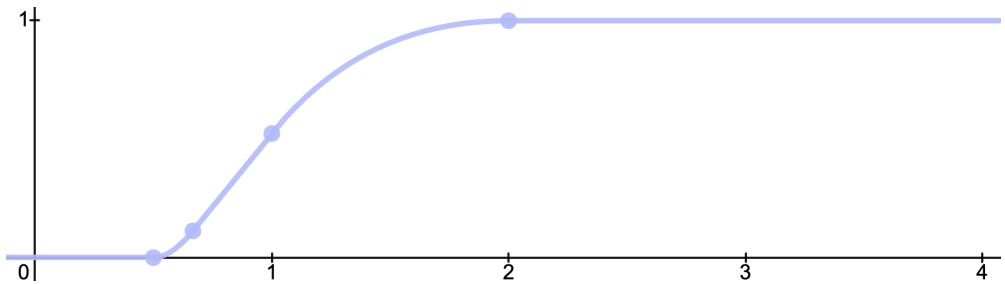
Set $R := V_a \backslash H_1$, $a > 0$. *Then*

$$\left\{ \frac{u_n^R}{s_n^R} \right\} = \left\{ \frac{(a_{j+1} - a)p_j + p_{j-1}}{(a_{j+1} - a)q_j + q_{j-1}} \;\middle|\; a_{j+1} > a \right\}$$

and

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \le k < n \mid \Theta(x, u_k^R/s_k^R) \le z\} = \bar{\mu}_R(S_z \cap R) \quad \text{a.s.}$$





Convergents/mediants of bounded approx. coeff.'s

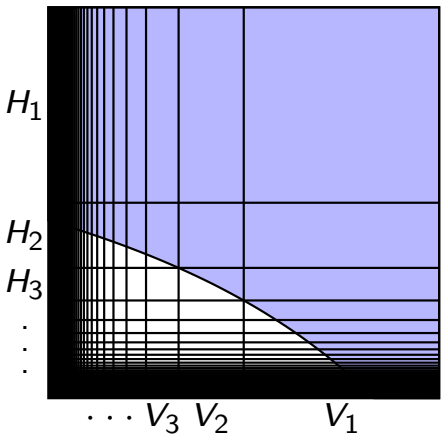
Theorem (Bosma 1990)

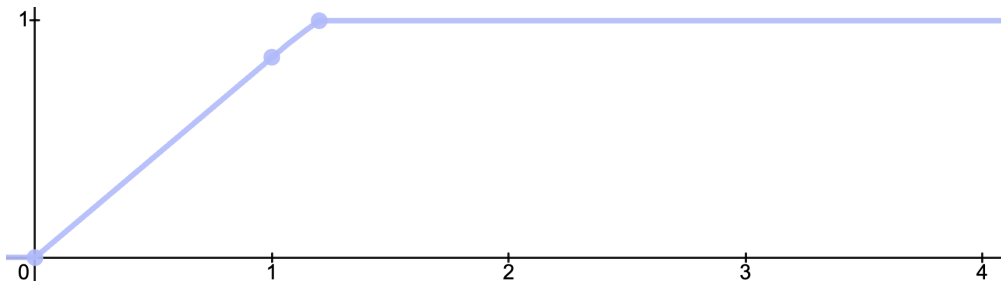
Set $R := S_{z_0}$, $z_0 \geq 0$. *Then*

$$\left\{ \frac{u_n^R}{s_n^R} \right\} = \left\{ \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}} \;\middle|\; \Theta\left(x, \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}}\right) \leq z_0 \right\}$$

and

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \leq k < n \mid \Theta(x, u_k^R/s_k^R) \leq z\} = \bar{\mu}_R(S_z \cap R) \quad \text{a.s.}$$





Convergents/λ^{th}-mediants, fixed digit a

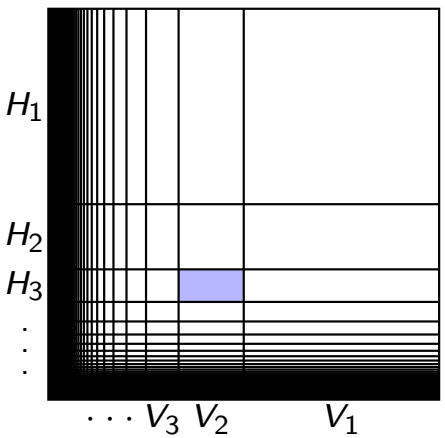
Theorem (Bosma 1990, $\lambda = a-1,\ a \geq 2$)

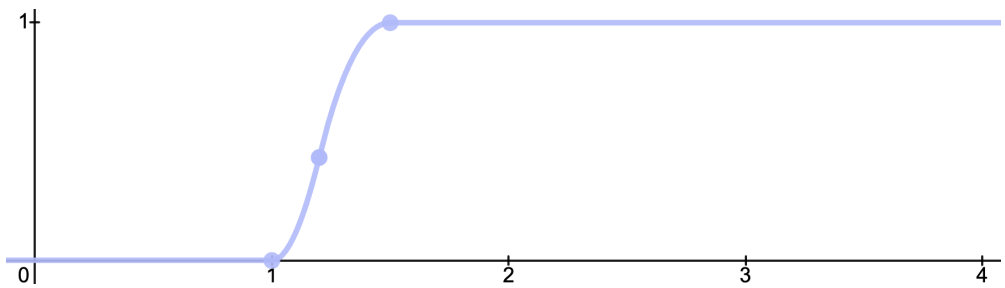
Set $R := V_{a-\lambda} \cap H_{\lambda+1},\ 0 \leq \lambda < a$. *Then*

$$\left\{ \frac{u_n^R}{s_n^R} \right\} = \left\{ \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}} \;\middle|\; a_{j+1} = a \right\}$$

and

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \leq k < n \mid \Theta(x, u_k^R/s_k^R) \leq z\} = \bar{\mu}_R(S_z \cap R) \quad \text{a.s.}$$





Convergents & first λ_0- & final a_0-mediants

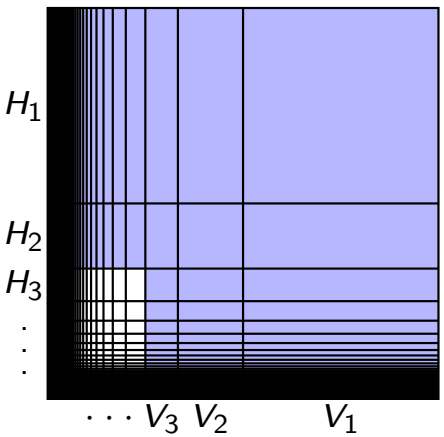
Theorem (Brown–Yin 1996, $\lambda_0 = a_0$; Ito '89, Bosma '90, Jager '91, $\lambda_0 = a_0 = 1$)

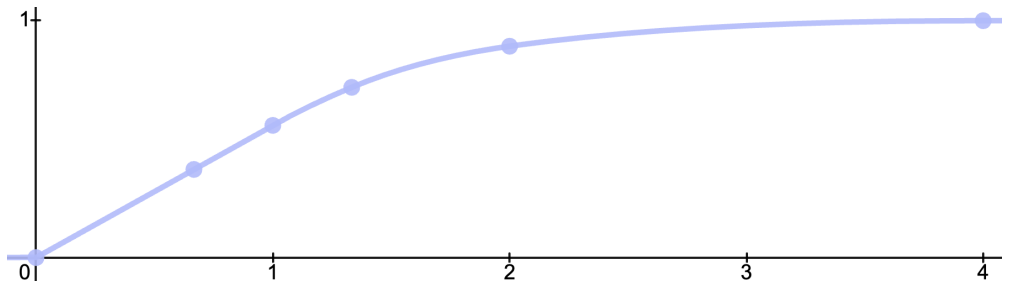
Set $R := \bigcup_{\lambda=0}^{\lambda_0} H_{\lambda+1} \cup \bigcup_{a=1}^{a_0} V_a$, $\lambda_0 \geq 0$, $a_0 \geq 1$. *Then*

$$\left\{ \frac{u_n^R}{s_n^R} \right\} = \left\{ \frac{\lambda p_j + p_{j-1}}{\lambda q_j + q_{j-1}} \;\middle|\; 0 \leq \lambda \leq \lambda_0,\ a_{j+1} > \lambda \right\} \cup \left\{ \frac{(a_{j+1} - a)p_j + p_{j-1}}{(a_{j+1} - a)q_j + q_{j-1}} \;\middle|\; 1 \leq a \leq a_0,\ a_{j+1} \geq a \right\}$$

and

$$\lim_{n\to\infty} \frac{1}{n} \#\{0 \leq k < n \mid \Theta(x, u_k^R/s_k^R) \leq z\} = \bar{\mu}_R(S_z \cap R) \quad \text{a.s.}$$





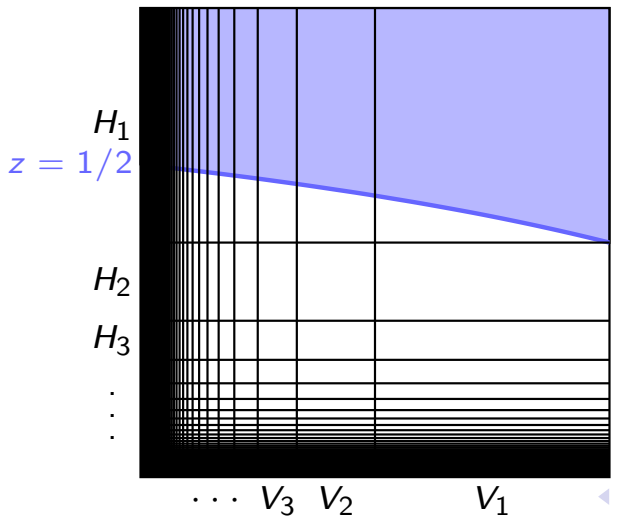
Legendre and Koksma's theorems

Throughout, we assume a weak Fatou–Grace ($\Theta(x, p/q) < 1 \implies p/q$ is a convergent or mediant).

Theorem (Legendre 1798)

If $\Theta(x, p/q) < 1/2$, then p/q is a convergent of x. Moreover, the constant $1/2$ is optimal.

Proof.



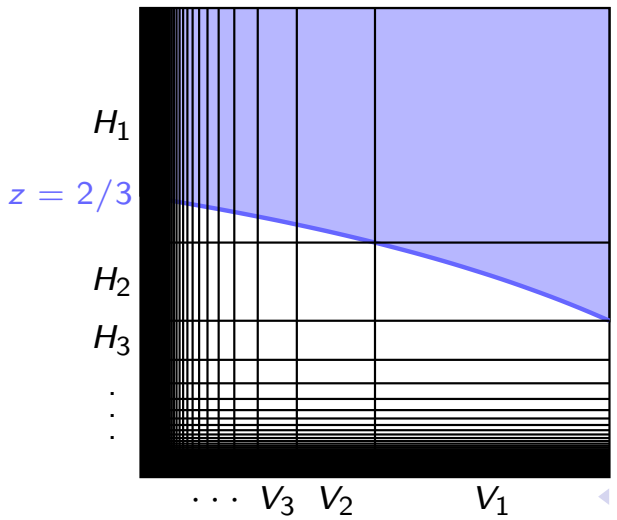
Legendre and Koksma's theorems

Throughout, we assume a weak Fatou–Grace ($\Theta(x, p/q) < 1 \implies p/q$ is a convergent or mediant).

Theorem (Koksma 1937/38)

If $\Theta(x, p/q) < 2/3$, then p/q is either a convergent or a first mediant convergent of x. The constant $2/3$ is optimal.

Proof.

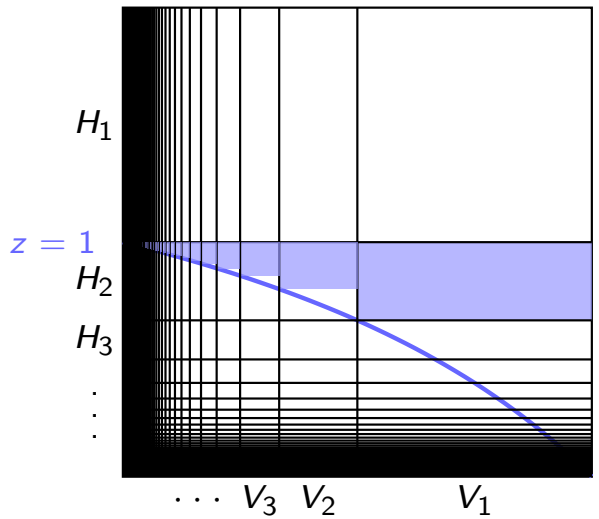


Kuipers–Meulenbeld

Theorem (Kuipers–Meulenbeld 1952)

Let $x = [0; a_1, a_2, \dots] \in (0,1)\backslash\mathbb{Q}$. *Suppose that* $a_{j+1} \geq 2$ *for some* $j \geq 0$. *If* $a_{j+1} \leq a_j + 1$, *then* $\Theta\left(x, \frac{p_j+p_{j-1}}{q_j+q_{j-1}}\right) < 1$.

Proof.



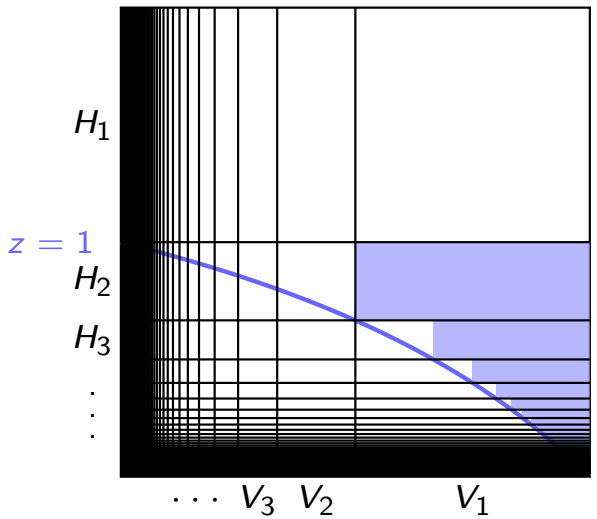
□

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Proof.



Signature of p/q

The *depth* of $p/q \in (0,1) \cap \mathbb{Q}$ is the length n of the shorter of its two expansions $p/q = [0; b_1, \ldots, b_n] = [0; b_1, \ldots, b_n - 1, 1]$, $b_n > 1$. The *signature* $\delta(x, p/q)$ of p/q with respect to $x \in (0,1)\backslash\mathbb{Q}$ is $(-1)^n \mathrm{sgn}(x - p/q)$.

Lemma

$\delta(x, u_n/s_n) = -1$ if and only if u_n/s_n is (i) a first mediant or (ii) a convergent with $a_{j_n - 1} = 1$, $j_n > 1$.

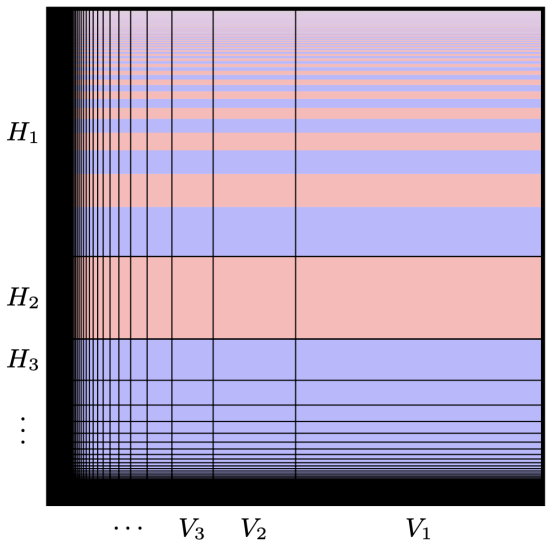


Figure: Red: $\delta = -1$.
Blue: $\delta = 1$.

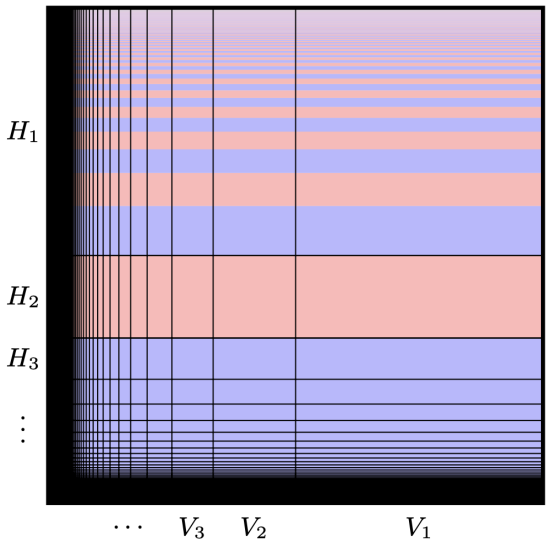
Barbolosi–Jager

Corollary (Barbolosi–Jager 1994)

Let $p/q \in \mathbb{Q} \cap (0,1)$ with $(p,q) = 1$ and $x \in (0,1)\backslash\mathbb{Q}$. Suppose $\delta(x, p/q) = 1$. Then

- *p/q is not a first mediant,*

Proof.



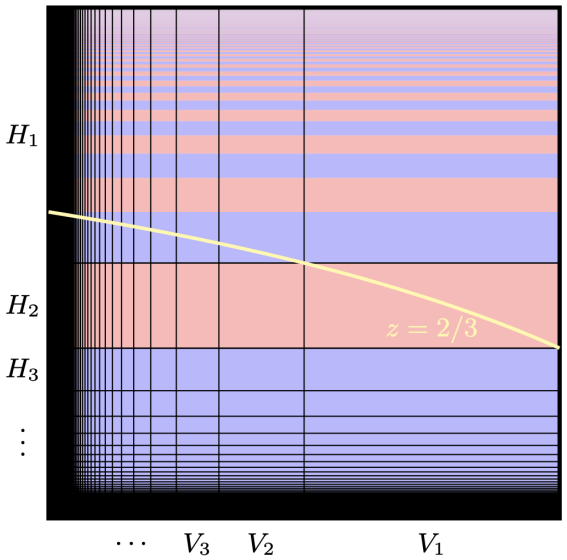
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- $\Theta(x, p/q) < 2/3 \implies p/q$ *is a convergent,*

Proof.



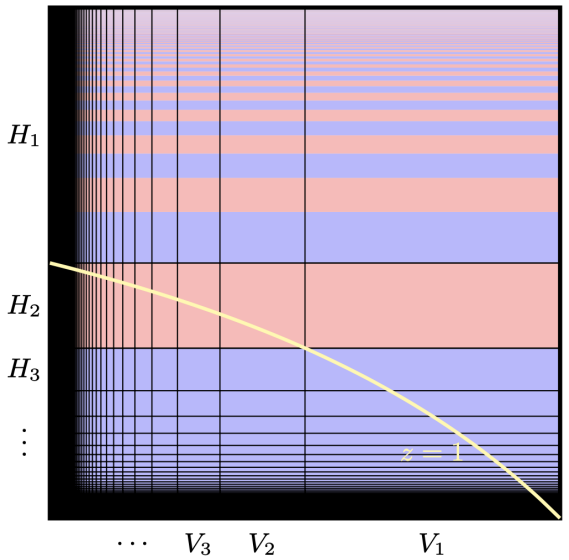
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- $\Theta(x, p/q) < 1 \implies p/q$ *is a convergent or a final mediant,*

Proof.



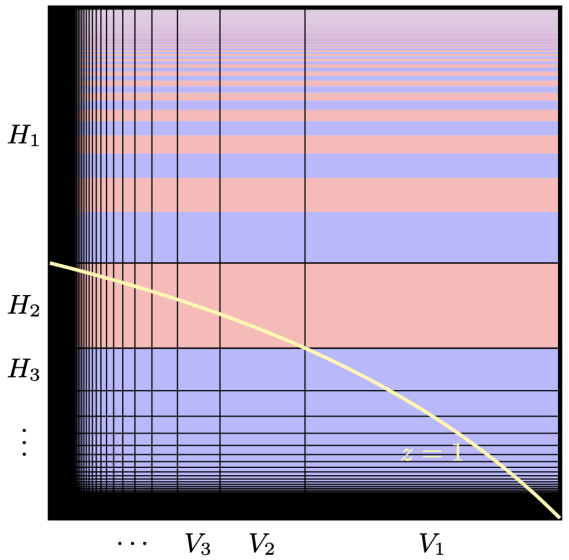
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- $\Theta(x, p/q) > 1 \implies p/q$ *is not a convergent,*

Proof.



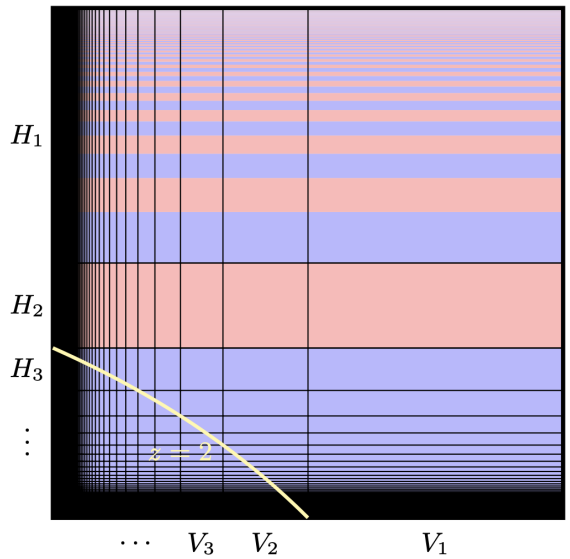
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- $\Theta(x, p/q) > 2 \implies p/q$ *is neither a convergent nor a nearest mediant.*

Proof.



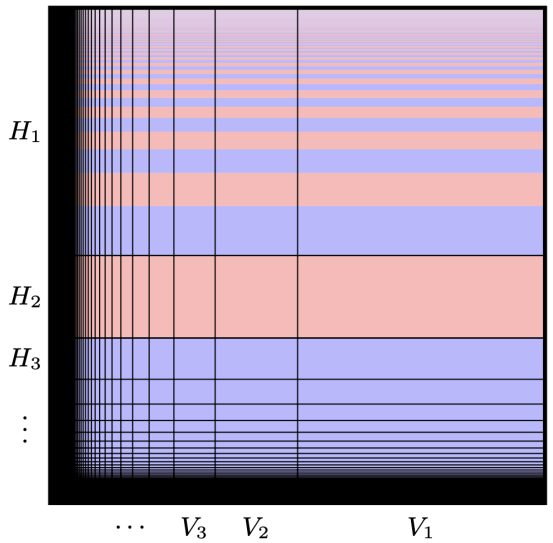
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- *if p/q is a final mediant of x, then it is also a first mediant,*

Proof.



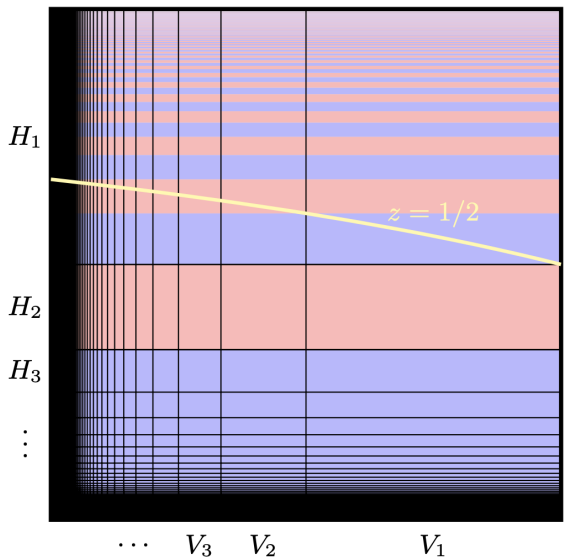
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Proof.



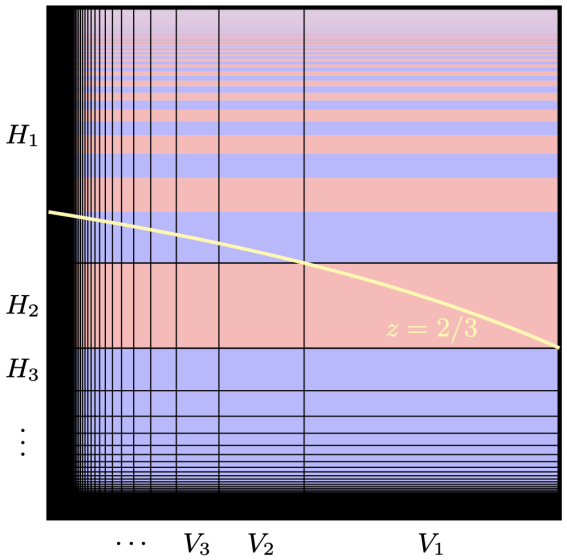
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Proof.



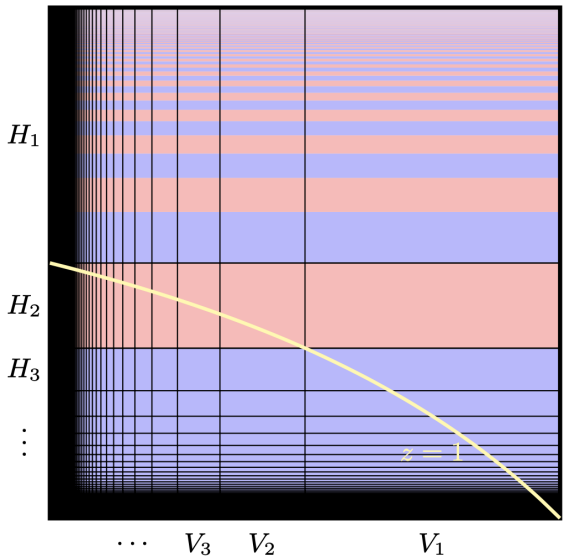
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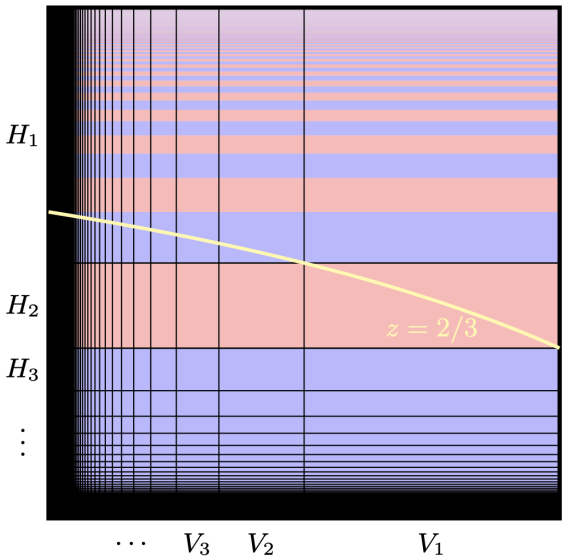
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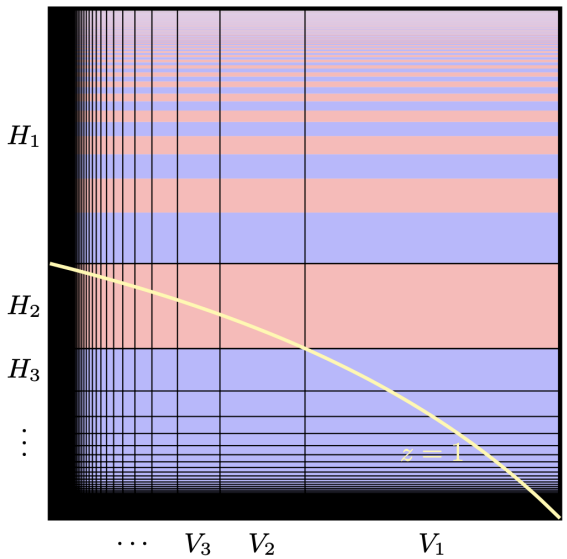
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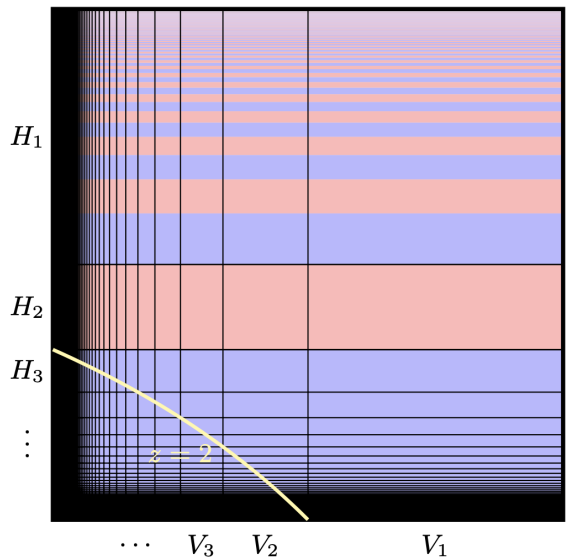
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